Sri Sai Charan Yarlagadda

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EDUCATION

Northeastern University Boston, MA | Sept 2024 – Dec 2026

Masters in Data Science, GPA: 3.73

Coursework: Natural Language Processing, Computer Vision, Supervised Machine Learning, Data Mining and Modeling, MLOps/LLMOps

Kalinga Institute of Industrial Technology Bhubaneswar, India | Sept 2020 – May 2024

B.Tech. in Computer Science Engineering, GPA: 3.76

Coursework: Machine Learning, Deep Learning, Data Analytics, Data Structures and Algorithms, Cloud Computing

EXPERIENCE

AI / ML Engineer Brooklyn, NY | Aug 2025 – Dec 2025

Crewasis.ai

- Extracted structured research metadata from Google Scholar via SerpAPI, collecting 10K+ publication records and ingesting them into S3 as JSON alongside 15+ domain-specific document corpora through automated Lambda-triggered preprocessing pipelines
- Fine-tuned Claude Sonnet 3.7 on the ingested domain corpus using QLoRA (4-bit NormalFloat) adapters via AWS Bedrock, specializing the model for enterprise document Q&A; and improving domain accuracy by 20% on a 500-sample held-out evaluation set
- Designed LangChain agentic workflows using ConversationalRetrievalChain for context-preserving multi-turn document queries and OutputFixingParser for structured JSON extraction, achieving 91.7% extraction accuracy across 2,000+ monthly documents and reducing manual review cycles by 40%
- Deployed Llama 3.3 70B (4-bit GPTQ quantized) on EC2 G5 instances for batch document summarization, processing 2,000+ docs/month with 30% lower inference latency through vLLM serving and KV-cache optimization
- Orchestrated the full pipeline on AWS with S3 for corpus storage, Lambda for event-driven ingestion, EC2 for GPU model serving, Bedrock for managed LLM inference, and CloudWatch for monitoring, improving reliability by 35% and reducing end-to-end latency by 45%

AI / ML Intern Ahmedabad, India | Jul 2023 – Nov 2023

BrainerHub Solutions LLP

- Developed a Bi-LSTM intent classification model using PyTorch on 10K+ labeled enterprise text samples with GloVe embeddings and attention-based pooling, achieving 85% F1-score and integrating it as an upstream intent router feeding into GPT-3.5 Turbo for response generation
- Built an enterprise chatbot for corporate document proofreading by wrapping GPT-3.5 Turbo via the OpenAI API with domain-adapted system prompts and structured output formatting, reducing manual proofreading time by 25% across 3 client workflows
- Deployed the chatbot pipeline on AWS EC2 with Docker containerization and GCP Cloud Run for auto-scaling, tracking model performance and prompt configurations with MLflow across 50K+ processed records

PROJECTS

InnovateAI – RAG Literature Review Engine Sept 2025 – Dec 2025

- Architected a production RAG system using LangChain RetrievalQA chains with FAISS vector store and BM25 sparse retrieval fused via Reciprocal Rank Fusion (RRF), achieving 0.87 MRR@10 and 91.7% Top-5 Recall on 1,200+ academic documents
- Implemented chunk-level re-ranking using a cross-encoder model (ms-marco-MiniLM-L6) to refine candidate passages before generation, improving answer relevance by 14% on a 300-query human-annotated test set
- Integrated Gemini 2.0 Flash with 8 specialized agent tools for multi-step reasoning and context-aware generation, reducing P95 query latency from 6.7s to 2.8s (58% reduction)
- Orchestrated 2 Airflow DAGs on GCP Cloud Composer for document ingestion and embedding refresh, containerized the model endpoint with Docker for GKE deployment, and tracked 15+ retrieval and latency metrics via MLflow and DVC

TheraBot – Emotion-Aware Mental Health AI Jan 2025 – Apr 2025

- Built a sequential NLP pipeline where BERTweet classifies user emotion across 7 categories (73% F1 on 25K+ samples), RoBERTa detects sarcasm to correct misclassified tone (74% F1), and distilGPT2 fine-tuned on Psychology 10K dataset with LoRA adapters generates empathetic responses (24% empathy uplift)
- Developed Flask REST API integrating Google Cloud Speech-to-Text API (RecognizeRequest with linear16 encoding) for real-time audio transcription, routing transcribed text through the emotion pipeline and delivering responses at 200ms P95 latency
- Implemented conversation memory using a sliding window of 5 prior exchanges stored in Redis, enabling the model to maintain therapeutic context across multi-turn sessions and improving user-rated coherence by 18% over stateless baselines

MultiMedAI – Multimodal Medical Intelligence Platform Oct 2024 – Dec 2024

- Built a multi-purpose medical AI platform combining image synthesis, diagnostic report generation, and visual question answering across radiology, dermatology, and pathology domains using 12K paired images and clinical reports
- Trained a Latent Diffusion Model (Stable Diffusion v1.5 with UNet denoiser and DDPM scheduler) using a Vision Transformer (ViT-L/14) text encoder for conditioning via HuggingFace diffusers, generating synthetic diagnostic images from clinical prompts with FID score of 42.3 and 89% clinician-rated plausibility
- Developed a medical VQA module using BiomedCLIP (ViT-B/16 backbone) for image encoding and a T5-base transformer decoder with cross-attention layers for answer generation, enabling clinicians to query images with natural language and receive structured diagnostic responses with 81% accuracy on 1,500 test cases
- Engineered a data preprocessing pipeline with torchvision transforms for multi-format ingestion (DICOM, PNG, TIFF), histogram equalization, and augmentation (rotation, flip, elastic deformation), expanding the limited 12K dataset to 36K effective training samples

TECHNICAL SKILLS

Languages: Python, SQL, R, C++, JavaScript

NLP and LLMs: HuggingFace Transformers, LangChain, Prompt Engineering, Fine-tuning (LoRA, QLoRA), RAG, vLLM, spaCy, NLTK

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost, HuggingFace diffusers, OpenCV, BiomedCLIP

Methods: Supervised and Unsupervised Learning, Deep Learning, Transfer Learning, Generative Modeling, Retrieval-Augmented Generation, Agentic Workflows

Cloud and MLOps: AWS (S3, Lambda, EC2, SageMaker, Bedrock), GCP (Cloud Run, Composer, GKE), Docker, Kubernetes, Git, CI/CD, MLflow, Airflow, DVC

Visualization: Matplotlib, Seaborn, Plotly, Tableau, Streamlit, Power BI